

- 系统

1. 量子统计: 密度矩阵: $\hat{\rho} = \exp[-\beta(\hat{H} - \mu_i \hat{N}_i)]$, 对系统中第 M_i 个系统: $P \propto \langle M_i | \hat{\rho} | M_i \rangle$

则有: 系统 $Z = \text{Tr}(\hat{\rho})$, 力学量平均 $\langle \hat{A} \rangle = \frac{\sum A_i P_i}{\sum P_i} = \frac{\text{Tr} \hat{A} \hat{\rho}}{\text{Tr} \hat{\rho}}$

$V \rightarrow \infty$ 时, 常用力学量: $P = \frac{\partial \ln Z}{\partial V}$

$$N_i = \frac{\partial \ln Z}{\partial \mu_i}$$

$$S = \frac{\partial \ln Z}{\partial T}$$

$$E = -PV + TS + \mu_i N_i$$

2. 单Boson自由度: (k 取 1)

平衡时可任意简并 $\Rightarrow$ 无法定义体积 $\Rightarrow$ 无压

粒子数 $n \rightarrow$ 态 $|n\rangle$ 构成规范正基底: $\langle n | m \rangle = \delta_{nm}$, $\sum_{n=0}^{\infty} |n\rangle \langle n| = 1$

产生算符 $a^\dagger, a$, $a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$, $a |n\rangle = \sqrt{n} |n-1\rangle \Rightarrow a^\dagger a |n\rangle = n |n\rangle, a^\dagger a |0\rangle = 0$

$$\hat{N} = a^\dagger a, |n\rangle = \frac{1}{\sqrt{n!}} a^\dagger |0\rangle, [a^\dagger, a] = 1$$

Hamilton: $H = \frac{1}{2} \omega [a^\dagger a + a a^\dagger] = \hat{N} \omega + \frac{\omega}{2}$

配分函数: $\hat{\rho} = e^{-\beta(H - \mu N)} = e^{-\beta(\omega - \mu)N} \cdot e^{-\beta \frac{\omega}{2}}$ 省略

$$Z = \text{Tr} \hat{\rho} = \sum_{n=0}^{\infty} \langle n | e^{-\beta(\omega - \mu)N} | n \rangle = \frac{1}{1 - e^{-\beta(\omega - \mu)}}, \langle N \rangle = \frac{\partial \ln Z}{\partial \beta(\omega - \mu)} = \frac{1}{e^{\beta(\omega - \mu)} - 1}$$

$$\langle E \rangle = \omega \langle N \rangle \quad (\mu \rightarrow \omega \text{ 会发散})$$

经典极限: $N \ll 1 \rightarrow T \ll \omega - \mu \rightarrow N = e^{-\beta(\omega - \mu)}$

量子极限: $N \gg 1 \rightarrow T \gg \omega - \mu \rightarrow N = \frac{1}{\beta(\omega - \mu)}$

3. 单Fermion 自由度: $a |1\rangle = |0\rangle, a^\dagger |0\rangle = |1\rangle, \underline{a |0\rangle = a^\dagger |1\rangle = 0}$

即: Grassman 代数 $aa = a^\dagger a^\dagger = 0, \hat{N} = a^\dagger a, \{a^\dagger, a\} = 1 \rightarrow \hat{H} = \frac{\omega}{2}(a^\dagger a - a a^\dagger) = \omega(\hat{N} - \frac{1}{2})$

$$Z = \sum_{n=0}^1 \langle n | \hat{\rho} | n \rangle = 1 + e^{-\beta(\omega - \mu)} \Rightarrow N = \frac{1}{e^{\beta(\omega - \mu)} + 1}$$

4. 无相互作用:

第*i*个组分 N_i, H_i, μ_i , 箱归一: 波长 $\lambda_i = 2L/j_i \Rightarrow d^3x = L \frac{d^3p_i}{(2\pi)^3}$

$$H = \sum H_i, \hat{N} = \sum N_i \Rightarrow Z = \text{Tr} e^{-\beta(H - \mu N)} = \prod_i \text{Tr} e^{-\beta(H_i - \mu_i N_i)} = \prod_i Z_i$$

$$\ln Z = \sum_{i_x, i_y, i_z} \ln Z_{i_x i_y i_z} \stackrel{L \rightarrow \infty}{\approx} V \int \frac{d^3p}{(2\pi)^3} \ln Z(p)$$

色散: $\omega = \omega(p)$

$$\text{e.g.: } \omega = \sqrt{p^2 + m^2} \rightarrow \ln Z = V \int \frac{d^3p}{(2\pi)^3} \ln(1 \pm e^{\beta(\omega - \mu)})^{\pm 1} \rightarrow P = \frac{m^4 T^4}{2\pi^2} e^{\frac{\mu}{T}} K_2\left(\frac{m}{T}\right)$$

时间反演性: 过程可逆 $\rightarrow$ 平衡状态、前后化学势相同)

二. 泛函积分

1. Boson 跃迁:

$$\text{场算符 } \hat{\phi}(x,0) |\phi\rangle, \int |\phi\rangle \langle \phi| d\phi(x) = 1, \langle \phi_a | \phi_b \rangle = \prod_x \delta(\phi_a(x) - \phi_b(x))$$

$$\hat{\pi}(x,0) |\pi\rangle, \int |\pi\rangle \langle \pi| \frac{d\pi(x)}{2\pi} = 1, \langle \pi_a | \pi_b \rangle = \prod_x \delta(\pi_a(x) - \pi_b(x))$$

$$\text{对易: } \langle x | p \rangle = e^{ipx} \rightarrow \langle \phi | \pi \rangle = e^{i \int d^3x \pi \phi}$$

$$H = \int d^3x \mathcal{H}(\pi, \phi) \Rightarrow \text{不显含时间时 } U(t) = e^{-iHt} \quad (\hbar=1) \quad \text{跃迁振幅 } \langle \phi_a | U | \phi_b \rangle = \langle \phi_a | e^{-iHt} | \phi_b \rangle$$

$$\text{分隔: } \Delta t = \frac{t}{N} \Rightarrow \langle \phi_b | U(t) | \phi_a \rangle = \langle \phi_b | \int \prod_i |\pi_i\rangle \langle \pi_i| e^{-iH \Delta t} | \phi_i\rangle d\pi_i d\phi_i \cdot |\phi_a\rangle$$

$$= \langle \phi_b | \int \prod_i d\pi_i d\phi_i |\pi_i\rangle \prod_i \langle \phi_{i+1} | \pi_i \rangle \langle \pi_i | e^{-iH_i \Delta t} | \phi_i \rangle \cdots \langle \phi_1 | \phi_a \rangle$$

$$= \int \prod_i d\pi_i d\phi_i \prod_i e^{-iH_i \Delta t + i \int d^3x \pi_i (\phi_{i+1} - \phi_i)} \langle \phi_b | \phi_a \rangle$$

$$= \int D\pi \int_{\phi(x,0)=\phi_a}^{\phi(x,t)=\phi_b} D\phi e^{-i \int_0^t H(\pi, \phi) dt - \pi_i(\phi_i)} = N \int D\phi \cdot e^{-i \int_0^t [T(\phi, \dot{\phi}) - V] dt}$$

$$\Rightarrow S_{ij} = \langle \phi_j | e^{-\frac{iHt}{\hbar}} | \phi_i \rangle$$

$$\text{周期性: } \langle \phi_a | U | \phi_a \rangle = \int D\lambda \int_{\phi_a}^{\phi_a} D\phi e^{-iS}$$

$$P_{ii} = \frac{(\text{Tr} S^\dagger (\text{Tr} S))}{(S_{ij}^\dagger S_{ij})}$$

2. 配分函数:

$$\mathcal{H}(\pi, \phi), \hat{N}(\pi, \phi) \Rightarrow Z = \text{Tr} e^{-\beta(H - \mu N)} = \sum_a \int \langle \phi_a | e^{-\beta(H - \mu N)} | \phi_a \rangle d\phi_a, \quad -\beta(H - \mu N) = -\beta \int d^3x (\hat{\mathcal{H}} - \mu \hat{N})$$

$$\text{虚时间 } \tau = it \Rightarrow Z = \int \langle \phi_a | \prod \int dx_i d\phi_i |x_i\rangle \langle x_i| e^{-\Delta\tau(H_i - \mu N_i)} |x_i\rangle \langle x_i| \dots | \phi_a \rangle$$

$$= \int_{\phi_0=\phi_a}^{\phi_\beta=\phi_a} D\phi \int D\pi e^{\int_0^\beta d\tau \int d^3x [i\pi \frac{\partial \phi}{\partial \tau} - \mathcal{H} + \mu N]}$$

→ $\phi(x,0)=\phi(x,\beta)$ 的所有场 (环路代数) ☆

概率性: $P_{aa} = \frac{\langle \phi_a | e^{\beta H + \beta \mu N} | \phi_a \rangle}{Z}$

↓

$d\beta \Rightarrow \langle \phi_0 | e^{\beta H + \beta \mu N} | \phi_a \rangle$

3. 实标量场: $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - U(\phi)$, $U(\phi) = g\phi^3 + \lambda\phi^4$

→ $\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \partial_0 \phi \Rightarrow \mathcal{H} = \frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2 + U(\phi)$

$Z_\beta = \int D\pi \int D\phi e^{S'}$, $S' = \int_0^\beta d\tau \int d^3x \mathcal{L} + \mu N + i\pi \frac{\partial \phi}{\partial \tau} - \pi \frac{\partial \phi}{\partial \tau}$ ← $\tau = i\tau$, 为0 $= \int_0^\beta d\tau \int d^3x \mathcal{L} + \mu N$

离散化分析: $Z = \lim_{n \rightarrow \infty} \left(\prod_{i=1}^n \int_{\phi_{i-1}}^{\phi_i} \frac{d\phi_i}{\sqrt{2\pi}} \right) e^{\sum_{i=1}^n \int d^3x [i\pi(\phi_{i1}-\phi_{i2}) - \Delta\tau (\frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 + U(\phi))]}$

↓ 空间划分: $V = L^3$, $L = Ma$, 取 $\pi_j = \frac{A_j}{(a^3 \Delta\tau)^{1/2}}$, 积 $A_j \Rightarrow \int_{-\infty}^{\infty} \frac{dA_j}{\sqrt{2\pi}} e^{-\frac{1}{2}A_j^2 + i(\frac{\partial}{\partial \tau})^{1/2}(\phi_{j1}-\phi_{j2})A_j} = \frac{1}{\sqrt{2\pi}} e^{\frac{-a^3(\phi_{j1}-\phi_{j2})^2}{2\Delta\tau}}$

$Z = \lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{2\pi}} \right)^n \prod_{i=1}^n \int d\phi_i e^{\sum_{i=1}^n \int d^3x \cdot \frac{1}{2}(\frac{\phi_{i1}-\phi_{i2}}{\Delta\tau})^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2 - U + \mu N}$

$= N' \int_{\phi_{i1}} D\phi e^{\int_0^\beta d\tau \int d^3x \mathcal{L}}$ 故 $Z = N' \int D\phi e^{\int_0^\beta d\tau \int d^3x \mathcal{L}}$

周期性条件: $\phi(x, \beta) = \phi(x, 0)$

对 $U(\phi)$, $U\phi = \phi \Rightarrow$

$S' = \int_0^\beta d\tau \int d^3x \mathcal{L} + \mu N$

无相互作用(实标量场 $\mu=0$) $S' = i \int_0^\beta d\tau \int d^3x (\partial_\mu \phi)^2 + m^2 \phi^2$, 对其 Fourier 展开: $\phi(x, \tau) = \frac{1}{\sqrt{V}} \sum_{\mathbf{p}} \sum_{\mathbf{n}} e^{(i\mathbf{p}\mathbf{x} + \omega_n \tau)} \phi_n(\mathbf{p})$, $\omega_n = \frac{2\pi n}{\beta}$, $\omega^2 = \mathbf{p}^2 + m^2$

↓ $= -\frac{1}{2} \int_0^\beta d\tau \int d^3x \phi (-\partial_\mu \partial^\mu + m^2) \phi$ $S' = -\frac{1}{2} \beta^3 \sum_{\mathbf{p}} \sum_{\mathbf{n}} (\omega_n^2 + m^2) \phi_n(\mathbf{p}) \phi_n^*(\mathbf{p})$

$Z = N' \cdot \prod_{\mathbf{p}} \prod_{\mathbf{n}} \left[\int_{-\infty}^{\infty} dA_n(\mathbf{p}) \cdot e^{-\frac{1}{2} \beta^3 (\omega_n^2 + m^2) A_n^2(\mathbf{p})} \right] = N' \prod_{\mathbf{p}} \prod_{\mathbf{n}} \frac{1}{\sqrt{2\pi}} \frac{1}{\beta^3 (\omega_n^2 + m^2)}$ $N' \int D\phi e^{-\frac{1}{2} \phi^T D \phi} = N'' (\det D)^{-\frac{1}{2}}$

即: $Z = N'' \cdot (\det D)^{-\frac{1}{2}}$ ← $\sum_{\mathbf{n}} \frac{1}{n^2 + (\frac{1}{2\pi})^2} = \frac{(2\pi)^2}{\theta} (1 + e^{-\theta})$

$\ln Z = -\frac{1}{2} \sum_{\mathbf{n}} \sum_{\mathbf{p}} \ln \beta^3 (\omega_n^2 + m^2) = -\frac{1}{2} \sum_{\mathbf{p}, \mathbf{n}} \ln [(2\pi n)^2 + \omega^2 \beta^2] = -\frac{1}{2} \sum_{\mathbf{p}, \mathbf{n}} \int_1^{\beta \omega^2} \frac{d\theta^2}{\theta^2 + (2\pi n)^2} + \ln [1 + (2\pi n)^2]$

$= -\sum_{\mathbf{p}, \mathbf{n}} \int_1^{\beta \omega^2} d\theta (\frac{1}{2} + \frac{1}{e^\theta - 1}) = V \int \frac{d^3p}{(2\pi)^3} [-\frac{1}{2} \beta \omega - \ln(1 - e^{-\beta \omega})]$

→ $E_0 = -\partial_\beta \ln Z_0 = -\frac{1}{2} V \int \frac{d^3p}{(2\pi)^3} \omega$, $P_0 = \frac{1}{\beta} \partial_V \ln Z_0 = -\frac{E_0}{V}$

4. BEC

复标量场: $\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m \phi^\dagger \phi - \lambda (\phi^\dagger \phi)^2$, 守恒荷: $e^{i\alpha}$

$$\hookrightarrow \mu \frac{\partial \mathcal{L}'}{\partial \alpha} = \frac{\partial \mathcal{L}'}{\partial \alpha} = 0 \Rightarrow j_\mu = \frac{\partial \mathcal{L}'}{\partial \alpha_\mu} = i(\phi^\dagger \partial_\mu \phi - \partial_\mu \phi^\dagger \phi) \Rightarrow \int j_0 dV = i(\phi^\dagger \pi - \pi^\dagger \phi)$$

分离实部虚部: $\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$, $\mathcal{H} = \frac{1}{2}[\pi_1^2 + \pi_2^2 + (\nabla\phi_1)^2 + (\nabla\phi_2)^2 + m^2\phi_1^2 + m^2\phi_2^2 + \mu]$

$$\hookrightarrow Z = \int D\phi_1 D\phi_2 \int_{\mathcal{M}} D\pi_1 D\pi_2 e^{\int d^4x [i\pi_1 \partial_t \phi_1 + i\pi_2 \partial_t \phi_2 - \mathcal{H} + \mu(\phi_1^2 + \phi_2^2)]} \Rightarrow \int d^4x [i\pi_1 \partial_t \phi_1 + i\pi_2 \partial_t \phi_2 - \mathcal{H} + \mu(\phi_1^2 + \phi_2^2)]$$

$$= \int_{\mathcal{M}} D\phi_1 D\phi_2 e^{\int d^4x (\mathcal{L} + \mu j_0)} \quad j_0 \text{ 不为 } \mu \phi^\dagger \phi \Rightarrow j_0 \text{ 动量依赖} \xrightarrow{\text{积分}} = (N')^2 \int_{\mathcal{M}} D\phi_1 D\phi_2 e^{\int d^4x [\frac{1}{2}(\frac{\partial \phi_1}{\partial t} - i\mu \phi_2)^2 - \frac{1}{2}(\frac{\partial \phi_2}{\partial t} + i\mu \phi_1)^2 - V]}$$

展开: $\lambda=0$ 时: $\phi_1 = \sqrt{2}\xi \cos\theta + \sqrt{\frac{V}{2}} \sum_{\mathbf{n}} \sum_{\mathbf{p}} e^{i(\mathbf{p}\cdot\mathbf{x} + \omega_{\mathbf{n}} t)} \phi_{1,\mathbf{n}}(\mathbf{p})$ $\rightarrow Z = (N')^2 \prod_{\mathbf{n}} \prod_{\mathbf{p}} \int d\phi_{1,\mathbf{n}}(\mathbf{p}) d\phi_{2,\mathbf{n}}(\mathbf{p}) e^{\mathcal{S}}$

$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi$ ($\omega_n = \sqrt{\mathbf{p}^2 + m^2}$) $\phi_2 = \sqrt{2}\xi \sin\theta + \sqrt{\frac{V}{2}} \sum_{\mathbf{n}} \sum_{\mathbf{p}} e^{i(\mathbf{p}\cdot\mathbf{x} + \omega_{\mathbf{n}} t)} \phi_{2,\mathbf{n}}(\mathbf{p})$ $\mathcal{S} = \beta V (\mu^2 - m^2) \xi^2 - \frac{1}{2} \sum_{\mathbf{n}} \sum_{\mathbf{p}} (\phi_{1,\mathbf{n}}(\mathbf{p}) \cdot \phi_{2,\mathbf{n}}(\mathbf{p})) D(\phi_{1,\mathbf{n}}(\mathbf{p}), \phi_{2,\mathbf{n}}(\mathbf{p}))$

$\phi_0 = \xi e^{i\theta}$ 为基态场 $\downarrow$ $\mathcal{L} = -\frac{1}{2}(\frac{\partial \phi_1}{\partial t} - i\mu \phi_2)^2 - \frac{1}{2}(\frac{\partial \phi_2}{\partial t} + i\mu \phi_1)^2 - V$ $\downarrow$ $D = \beta^2 \begin{pmatrix} \omega^2 + \omega_0^2 - m^2 & -2\mu\omega_0 \\ 2\mu\omega_0 & \omega^2 + \omega_0^2 - m^2 \end{pmatrix}$

$\hookrightarrow \phi_1(0) = \phi_2(0) = 0$

$|\mu| = m$ 时: $\rho = \frac{T}{V} (\partial_\mu \ln Z)_{\mu=m} = 2m\xi^2 + \rho^*(\beta, \mu=m)$ $\ln Z = \beta V (\mu^2 - m^2) \xi^2 + \ln(\det D)^{\frac{1}{2}}$

$\rho^* = \int \frac{d^3p}{(2\pi)^3} \frac{1}{e^{\beta(\omega - \mu)} - 1} - \frac{1}{e^{\beta(\omega + m)} - 1} \Rightarrow \xi^2 = \sqrt{\frac{\rho - \rho^*(\beta, \mu=m)}{2m}}$ (待求量) $= \rho V (\mu^2 - m^2) \xi^2 - \frac{1}{2} \sum_{\mathbf{n}, \mathbf{p}} \ln[\beta^2 (\omega_n^2 + (\omega - \mu)^2)] - \frac{1}{2} \sum_{\mathbf{n}, \mathbf{p}} \ln[\beta^2 (\omega_n^2 + (\omega + m)^2)]$

非相对论 $\rho^* \approx \int \frac{d^3p}{(2\pi)^3} \frac{1}{e^{\beta \frac{p^2}{2m}} - 1} + \frac{1}{e^{\beta m} - 1} \approx (\frac{mT}{2\pi})^{\frac{3}{2}} \xi^2 (\frac{1}{2})$ $= \rho V (\mu^2 - m^2) \xi^2 - V \int \frac{d^3p}{(2\pi)^3} [\beta \omega + \ln(1 - e^{-\beta(\omega - \mu)})] + \ln(1 - e^{-\beta(\omega + m)})]$

$\hookrightarrow \mu = m, \beta = \beta_c \Rightarrow T_c = \frac{m^2}{\pi} (\xi^2)^{\frac{2}{3}}$

相对论: $\approx \frac{mT^2}{6\pi^2} \rightarrow T_c = (\frac{3\rho}{\pi})^{\frac{1}{3}}$ $m \rightarrow 0$ 时 $k \rightarrow \infty$ 即: 全费米子 ($m=0$)

$\hookrightarrow$ 1). $|\mu| \leq m$ 才收敛 2) 与 θ 关系: $U(1)$ 对称

与非费米子: $\frac{\partial \ln Z}{\partial \xi} = 0 = 2\beta V (\mu^2 - m^2) \xi = 0$

5. 费米子 (旋)

$$\psi = \frac{1}{\sqrt{V}} \sum_{\mathbf{p}} \sum_{\mathbf{s}} \sqrt{\frac{m}{E}} [b(\mathbf{p}, \mathbf{s}) u(\mathbf{p}, \mathbf{s}) e^{-i\mathbf{p}\cdot\mathbf{x}} + d^\dagger(\mathbf{p}, \mathbf{s}) v(\mathbf{p}, \mathbf{s}) e^{i\mathbf{p}\cdot\mathbf{x}}], \quad \mathbf{p}\cdot\mathbf{x} = E t - \mathbf{p}\cdot\mathbf{x}$$

归一化: $\int d^3x \psi^\dagger \psi = 1 \Rightarrow \sum_{\mathbf{p}} \sum_{\mathbf{s}} |b(\mathbf{p}, \mathbf{s})|^2 + |d(\mathbf{p}, \mathbf{s})|^2 = 1$

$\mathcal{L} = \bar{\psi}(i\partial - m)\psi \rightarrow \mathcal{L} = \psi^\dagger \gamma^0 (i\gamma^0 \partial_t + i\gamma^i \partial_i - m)\psi$, 相对论性 $U(1) \rightarrow e^{i\alpha} \Rightarrow j^\mu = \psi^\dagger \gamma^0 \gamma^\mu \psi$, $Q = \int d^3x j^0 = \int d^3x \psi^\dagger \psi = 1$

动量 $\pi = i\psi^\dagger \Rightarrow \mathcal{H} = \bar{\psi}(-i\gamma^i \partial_i + m)\psi$

配分函数 $Z = \text{Tr} e^{-\beta(\mathcal{H} - \mu Q)}$

$$= \int D\psi D\psi^\dagger e^{\int d^4x \bar{\psi}(-i\gamma^0 \partial_t + i\gamma^i \partial_i - m + \mu\gamma^0)\psi}$$

性质: $\int d\eta_1 d\eta_2 \dots d\eta_1 d\eta_2 e^{\eta^T D \eta} = \det D$

$$\text{展开: } \psi_n(x, \tau) = \frac{1}{\sqrt{V}} \sum_{\mathbf{p}} e^{i(\mathbf{p} \cdot \mathbf{x} + \omega_n \tau)} \tilde{\psi}_{n, \mathbf{p}}(\mathbf{p})$$

格林函数:

$$\text{玻色: } G_B(x, y; \tau_1, \tau_2) = Z^{-1} \text{Tr}[\hat{\rho} T_\tau[\hat{\phi}(x, \tau_1) \hat{\phi}(y, \tau_2)]] \quad , \quad \hat{\phi}(x, \tau) = e^{\beta K} \hat{\phi}(x, 0) e^{-\beta K} \quad ; \quad \text{取 } \hat{\rho} = e^{-\beta K} \quad , \quad K = H - \mu Q$$

$\hat{\rho}$ 与 T_τ 对易

$$T_\tau(\phi(x, \tau_1), \phi(y, \tau_2)) = \theta(\tau_2 - \tau_1) \phi(x, \tau_2) \phi(y, \tau_1) + \theta(\tau_1 - \tau_2) \phi(y, \tau_2) \phi(x, \tau_1)$$

$$G_B(x, y; \tau, 0) = Z^{-1} \text{Tr}[e^{-\beta K} \phi(x, \tau) \phi(y, 0)] = Z^{-1} \text{Tr}[\phi(y, 0) e^{-\beta K} \phi(x, \tau)] = Z^{-1} \text{Tr}[e^{-\beta K} \phi(y, \beta) \phi(x, \tau)] \Rightarrow \phi(x, 0) = \phi(x, \beta)$$

迹轮换

$$\text{费米子: } T_\tau(\psi(x, \tau_1), \psi(y, \tau_2)) = \theta(\tau_1 - \tau_2) \psi(x, \tau_2) \psi(y, \tau_1) - \theta(\tau_2 - \tau_1) \psi(y, \tau_2) \psi(x, \tau_1)$$

$$G_F(x, y; \tau, 0) = Z^{-1} \text{Tr}[\hat{\rho} \psi(x, \tau) \psi(y, 0)] = Z^{-1} \text{Tr}[\hat{\rho} \psi(y, \beta) \psi(x, \tau)] = -G_F(x, y; \tau, \beta)$$

$$\text{即: } \psi(y, 0) = -\psi(y, \beta) \Rightarrow \omega_n = (2n+1)\pi T$$

将展开式代入Z:

$$Z = \left[\prod_n \prod_{\mathbf{p}} \int d\tilde{\psi}_{n, \mathbf{p}}^\dagger d\tilde{\psi}_{n, \mathbf{p}} \right] e^S \quad , \quad S = \sum_{n, \mathbf{p}} i \tilde{\psi}_{n, \mathbf{p}}^\dagger D_{n, \mathbf{p}} \tilde{\psi}_{n, \mathbf{p}} \quad , \quad D = -i\beta[(i\omega_n + \mu) - \gamma^0 \gamma \cdot \mathbf{p} - m\gamma^0]$$

↓

$$Z = \det D \Rightarrow \ln \det D = \text{Tr} \ln D = 2 \sum_n \sum_{\mathbf{p}} \ln [\beta^2 (\omega_n^2 + \mu^2 + \mathbf{p}^2)]$$

$$= \sum_n \sum_{\mathbf{p}} \ln \beta^2 (\omega_n^2 + (\omega_n + \mu)^2) + \ln \beta^2 \omega_n^2 (\omega_n - \mu)^2$$

$$\text{步骤同Boson: } \ln Z = 2V \int \frac{d^3 p}{(2\pi)^3} [\beta \omega + \ln(1 + e^{\beta(\omega - \mu)}) + \ln(1 + e^{\beta(\omega + \mu)})]$$

$$\text{总结: } \text{配分函数的路径积分表述: } Z = \text{Tr} e^{-\beta(H - \mu N)} = \sum_n \langle \phi_n | e^{-\beta(H - \mu N)} | \phi_n \rangle = \int_{\text{cir}} \mathcal{D}\phi \int \mathcal{D}\lambda e^{\int_0^\beta \int d^3x \cdot \pi \partial_t \phi - H + \mu N} = \int_{\text{cir}} \mathcal{D}\phi e^{\int_0^\beta \int d^3x \mathcal{L} + \mu N}$$

$$\text{对cir的边界条件: } \phi(x, \tau) \Rightarrow \text{玻色子: } \langle \phi_a \phi_b \rangle = \langle \phi_b \phi_a \rangle \Rightarrow G_B(x, y; \tau_1, \tau_2) = Z^{-1} \text{Tr}[\hat{\rho} T_\tau[\phi(x, \tau_1) \phi(y, \tau_2)]]$$

$$\text{Green: } \hat{\rho} = e^{-\beta K} \quad , \quad K = H - \mu Q \quad , \quad \langle \hat{A} \rangle = \frac{1}{Z} \text{Tr}(\hat{\rho} \hat{A}) \rightarrow G^{(1)}(x_1, x_2; \tau_1, \tau_2) = \langle T_\tau(\phi(x_1, \tau_1), \phi(x_2, \tau_2)) \rangle$$

$$\text{推广: } G^{(1)}(x, x_2; \tau_1, \tau_2) = \frac{1}{Z} \text{Tr}[\hat{\rho} T_\tau[\phi(x, \tau_1) \phi(x_2, \tau_2)]] \rightarrow G(x, y; \tau, 0) = \frac{1}{Z} \text{Tr}[\hat{\rho} T_\tau[\phi(x, \tau) \phi(y, 0)]]$$

$$\text{迹轮换: } \text{Tr}[A \cdots BC] = \text{Tr}[CA \cdots B] \Rightarrow G(x, y; \tau, 0) = \frac{1}{Z} \text{Tr}[\phi(y, 0) \hat{\rho} \phi(x, \tau)] = \frac{1}{Z} \text{Tr}[\hat{\rho} (\hat{\rho}^{-1} \phi(y, 0) \hat{\rho}) \phi(x, \tau)]$$

$$\hat{\rho} = e^{-iH\tau} \Rightarrow \hat{\rho}^{-1} \phi(y, 0) \hat{\rho} = \phi(y, \beta)$$

$\hat{\rho}$ 既作 密度, 又作 演化符? $\square$

$$\text{Boson: } T_c[\phi(x_1)\phi(x_2)] = \theta(t_2-t_1)\phi_2\phi_1 + \theta(t_1-t_2)\phi_1\phi_2 \Rightarrow G_B(x,y;\tau,0) = \frac{1}{Z} \text{Tr}[\hat{\rho} \phi(y,\beta)\phi(x,\tau)] = \frac{1}{Z} \text{Tr}[\hat{\rho} T_c(\phi(x,\tau)\phi(y,\beta))]$$

$$\text{即: air 为: } \phi(y,\beta) = \phi(y,0)$$

$$\text{Fermion: } T_c[\psi(x_1)\psi(x_2)] = \theta(t_2-t_1)\psi_2\psi_1 - \theta(t_1-t_2)\psi_1\psi_2 \Rightarrow G_F(x,y;\tau,0) = -G_F(x,y;\tau,\beta) \quad (\text{可得 } G_F(x,y;\tau_a,\tau_b) = -G_F(x,y;\tau_a,\beta-\tau_b))$$

$$\text{即: air 为 } \phi(y,\beta) = -\phi(y,0)$$

$$G_F(x,y;\tau,\beta) = 0$$

$$\text{自由子: Boson: } L = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi, \text{ 算子: } \phi_{(\tau,p)} = \frac{1}{\sqrt{V}} \sum_n e^{iP \cdot x + i\omega_n \tau} \phi_n(x)$$

$$\text{积分: } P = \frac{2\pi A_j (n+1)}{L}, A_j \text{ 是面积} \rightarrow \int d^3A_j = V \int \frac{d^3P}{(2\pi)^3} \rightarrow \sum_n \sum_{A_j} = \frac{V}{\beta} \int \frac{d^3P}{(2\pi)^3}, \phi^\dagger \phi = \sum_{p,n} \phi_{(p,n)}^\dagger \phi_{(p,n)}$$

$$\text{算符: } L = \phi^\dagger D \phi, \phi_{(p)} D \phi_{(p)} = \frac{\beta}{V} (-P^2 - m^2 - \omega_n^2) \phi_n^\dagger \phi_n \rightarrow S = \int_0^\beta d\tau \int d^3x L = \phi^\dagger [-\frac{1}{2}\beta^2(\omega^2 + \omega_n^2)] \phi = \frac{1}{2}(\partial_\mu \phi^\dagger - m^2 \phi)$$

$$Z = N \int D\phi e^S = N \prod_p \int d\phi_{(p)} e^S, S = -\frac{1}{2}\beta^2(\omega^2 + \omega_n^2) A_n^2$$

$$\text{故 } Z = N' \left(\frac{1}{\sqrt{2\pi}}\right)^n \sqrt{\det D}^{-1} = \prod_{p,n} \frac{1}{\sqrt{\beta^2(\omega^2 + \omega_n^2)}}, \ln Z = -\frac{1}{2} \left(\sum \ln(\beta^2 \omega^2 + (2\pi n)^2) \right) = V \int \frac{d^3P}{(2\pi)^3} \left(-\frac{1}{2}\beta \omega - \ln(1 - e^{-\beta \omega}) \right)$$

$$P = T \frac{\partial \ln Z}{\partial V}, E = -\partial_\beta \ln Z, N = T \frac{\partial \ln Z}{\partial \mu}$$

$$\text{Fermion: 同上, } L = \bar{\psi}(i\not{\partial} - m)\psi, \omega_n = \frac{2\pi(n+1)}{\beta} \Rightarrow \psi_{(\tau,p)} = \frac{1}{\sqrt{V}} \sum_n \sum_p e^{i(P \cdot x + \omega_n \tau)} \psi_{n,p}, L = \bar{\psi}^\dagger D \bar{\psi}$$

此处 $\int_0^\beta d\tau e^{i\omega_n \tau} = 0$ (反对称)
故无 β 项

$$L = \bar{\psi}(i\not{P} - m)\psi \Rightarrow L = \sum_n \sum_p e^{i(\Delta P x + \omega_n \tau)} \frac{1}{V} \bar{\psi}_{n,p}^\dagger \gamma^0 (i\not{\gamma} \cdot \vec{r} - m) \tilde{\psi}_{n,p} \Rightarrow S = \int_0^\beta d\tau \int d^3x L = \int_0^\beta d\tau \int d^3x \frac{L}{V}$$

$$L = -\bar{\psi} \gamma^0 \frac{\partial}{\partial \tau} \psi - \bar{\psi} (-i\not{\gamma} \cdot \vec{\nabla} + m) \psi - m \bar{\psi} \psi \Rightarrow D = -i\beta [-i(\omega_n + m) - \not{\gamma}^0 \not{\gamma} \cdot \vec{P} - m\gamma^0], Z = \det D$$

$$\text{动量: } G_P(P_1, P_2,$$